

SHREYASH GAURAV

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Education

National Institute of Technology, Rourkela

Master of Technology in Computer Science (CGPA - 8.45)

2024 – 2026

Rourkela, Odisha

Institute of Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur, CG

Bachelor of Technology in Computer Science And Engineering (CGPA - 9.31)

2020 – 2024

Bilaspur, Chhattisgarh

Work Experience

Samsung Research Institute Bangalore - SRIB

Research Intern, Platform Intelligence Team

Jan 2026 – Jul 2026

Bangalore

- Improved image-based NER F1 from **0.45 to 0.70 (+55%)** by fine-tuning Qwen3-VL-2B with LoRA, matching 8B-model quality at **~4x lower inference cost**.
- Built a 3-agent OCR + NER pipeline (LangGraph, Qwen3-VL, PaddleOCR v5) extracting and classifying text from real-world images across **13+ languages**, to drive downstream translate-vs-transliterate decisions.
- Eliminated manual annotation by distilling labels from the 8B model; validated and stratified ~20K images into an 80/10/10 split. Demoed the system at **Tech Glanz**, Samsung's internal tech summit.

Projects

Duplicate Image Detection Engine

C++20, CMake, libjpeg-turbo, scikit-learn, GoogleTest

July 2026 - Present

[GitHub](#)

- Built a C++20 near-duplicate image engine: perceptual hashing, SIMD Hamming matching, union-find clustering.
- Trained a calibrated classifier over 9 hash/image features: 95.8% precision, 96.7% recall on 1,200 labeled pairs.
- Zero-dependency decoding via libjpeg-turbo scaled DCT and a binary hash cache; 52 tests, CI with sanitizers.

Multimodal RAG System

Python, SBERT, CLIP, ChromaDB, GPT-4o

March 2026 - April 2026

[GitHub](#)

- Built an multi-modal RAG pipeline supporting text and image modalities.
- Used Sentence-BERT and CLIP embeddings stored in ChromaDB for unified vector search.
- Integrated GPT-4o for multimodal answer generation with source citation across both modalities.

Technical Skills

- Languages:** C++, Python, Java, C, JavaScript, HTML, CSS
- Data Science & ML:** PyTorch, scikit-learn, NumPy, Pandas, OpenCV
- Web Frameworks:** Node.js, Express.js, React, FastAPI
- Database & ODM:** MySQL, MongoDB, Mongoose
- Operating Systems:** Windows, Linux

Relevant Coursework

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|---------------------------------|-------------------------------|-----------------------|
| • Deep Learning | • Natural Language Processing | • Database Management |
| • Data Structures and Algorithm | • Operating System | • Computer Networks |

Publications

Classification of Plant Disease using a State-of-the Art Deep Learning Algorithm on a Tesla GPU

International Journal of Computing, 23(2), 2024

- Manjit Jaiswal, ..., **Shreyash Gaurav** et al.
- DOI: 10.47839/ijc.23.2.3542

Multi-Scale Diffusion Enhancement for Graph Clustering with Heat and PPR Kernels

ACM CODS-COMAD 2025, Young Researchers Symposium Track

- Devansh Maru, **Shreyash Gaurav**, Arpit Kurmi, Shashwat Panigrahi, Panthadeep Bhattacharjee
- URL: ikdd.acm.org/cods-2025/accepted-papers.php